

Resilience of Networks in the B5G Era

Toward dynamic assessment of availability with Digital Twins

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Acknowledging the work of a whole team: current and past members

Current members from Orange team:

- Guillaume FRAYSSE, PhD
- Bertrand DECOCQ, PhD
- Benoit LEMOINE
- Firas MAALEJ, intern
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Past members:

- Rui LI, PhD
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- Manelle NOUAR



Acknowledging the work of a whole team: collaborations

Risk and Resilience of Complex Systems (RRSC)
chaire at CentraleSupélec:

- Anne BARROS
- Zhiguo ZENG
- Yiping FANG

Collaborative project:

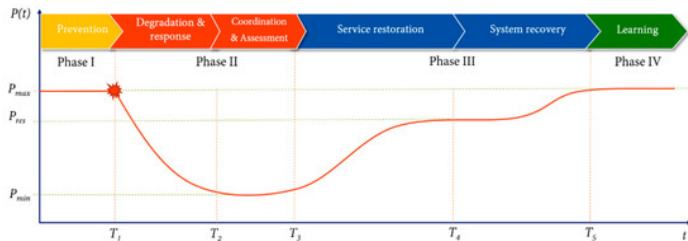
- SNS-JU Programmable Security Plane for 6G Ecosystems (MARE)



Definition(s)

Resilience of networks (from ResiliNets project <https://resilinetts.org>)

Definition: *Resilience* is the ability of the network to face various incidents, maintain an acceptable level of service and return to normal operation.



From Nait Belaid et al [1]

How do we measure resilience?

Resilience is a **multidimensional property**.

Availability

A

Is the service operational ?

Uptime / downtime

Reliability

MTBF

How often does it fail?

MTBF / MTTF

Recoverability

MTTR

How fast can it be restored?

MTTR / TTR

Service continuity

QoS

How much does service degrade?

Latency / loss / throughput

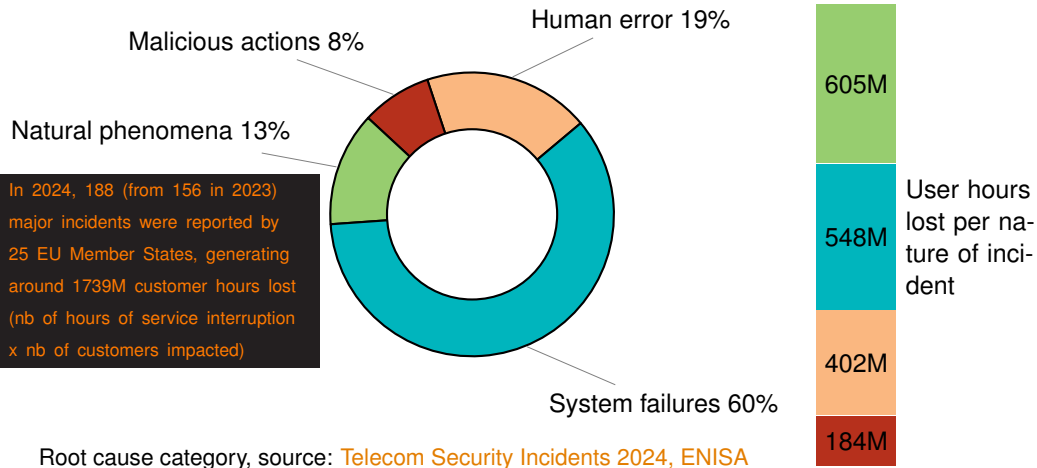
**Resilience
assessment**



$$A = \frac{\text{Uptime}}{\text{Observation time}}$$

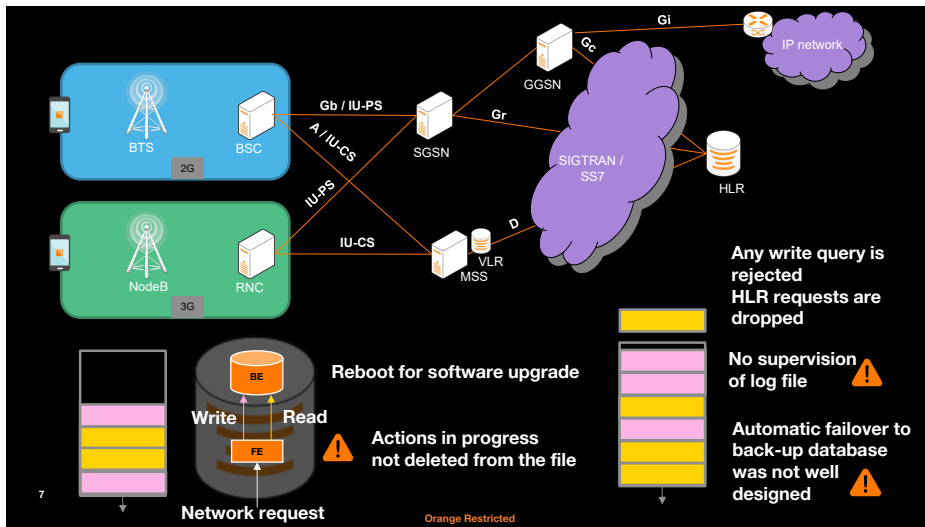


Root Causes of incidents and their impact (2024)



YoY evolution of incidents





Signaling storm : EE 4G network - March,2014

The Telegraph

Home News World Sport Finance Comment Culture Travel Life Women Fashion Technology Tech

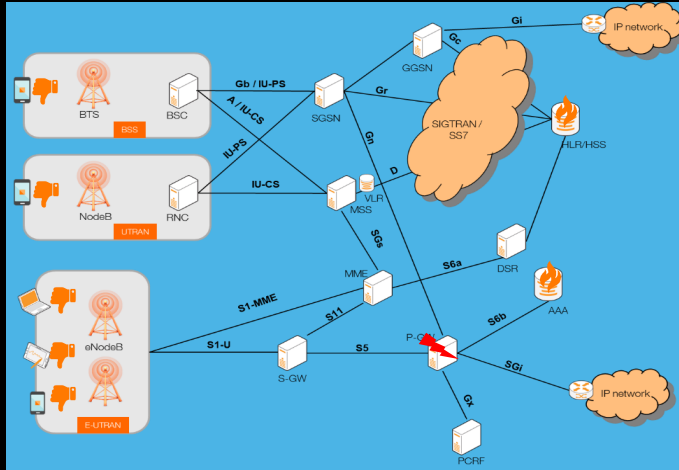
Home » Technology » Telecommunications » Technology Review » Video game » Technology Video

EE apologises for mobile network outage

EE blames for network is now back up and running, following an outage overnight

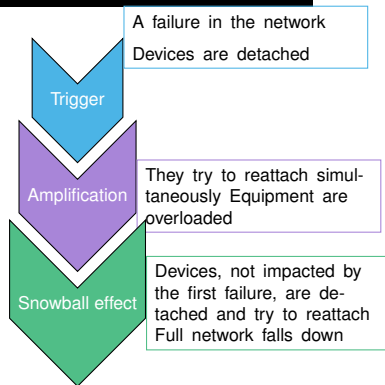


© 2014 EE and Orange UK, merged together in 2010 to make EE

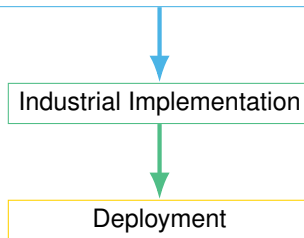


Redundancy and diversity were not sufficient anymore in 4G

Since 4G deployment, we face complex outages because of always-on devices



4G standards improvement / coordination mechanism



Lessons learned for 5G standards design

Resilience improvement for 5G

Redundancy/diversity

1

User Plan

New mechanisms allow a real user plan redundancy from an E2E perspective, combining access and core mechanisms

Redundancy/diversity

2

NF sets

NF (Network Function) can be organized as group of instances to allow load balancing, to react in case of failure of one instance

Load/Overload control

3

NF informs

Local or global mechanisms can be launched by an overloaded NF alerting its "neighbors" about its situation : timers, prioritization,...

4

Service continuity

Stateless NF

States information are stored in a separated database. In case of NF failure, another can take the floor retrieving the information from the database. No impact on the service

5

Dynamicity

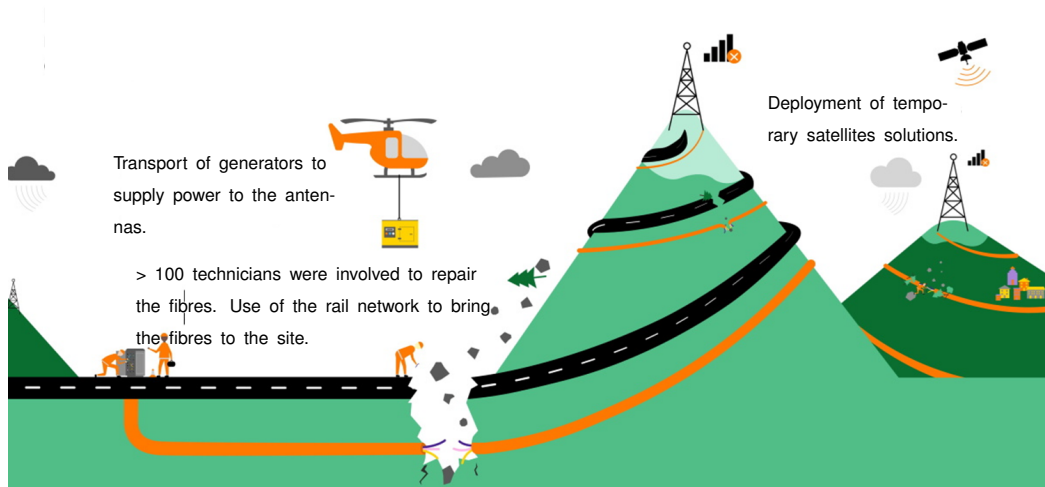
Self healing...

Dynamicity is ensured by restoration mechanisms at 5G level and/or by NF instantiation on the fly at virtual machines or containers level

Flood in la vallée de la Vésubie, October 2020



Flood in la vallée de la Vésubie, October 2020



New sources of problems

New problems for resilience

- More layers
- More dependencies
- More dynamicity
- More heterogeneity



New, and future, problems



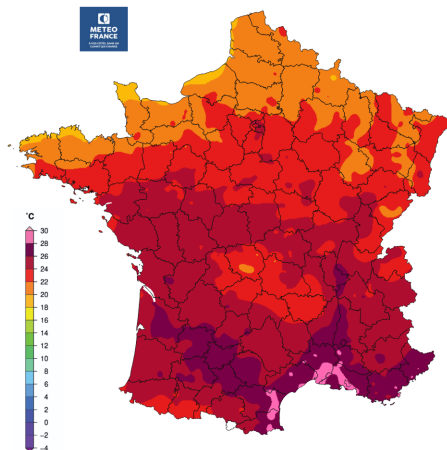
Mayotte, a French department in the Indian Ocean, was hit by tropical cyclone Chido on December 14th, 2024.

51 out of 54 mobile network antennas are out of service, mainly due to a lack of power. **99% of fixed-line Internet customers** are disconnected.

Increased complexity for the coordination of emergency services.



New, and future, problems



July 2026 is the hottest month ever recorded in France, average temperature was **24.9°C**. Climate change leads to higher average temperatures, longer heat waves, increased risks of fires, . . . Impact on material, infrastructure, people, ...

Global warming is not an hoax



PLANÈTE • RÉCHAUFFEMENT CLIMATIQUE

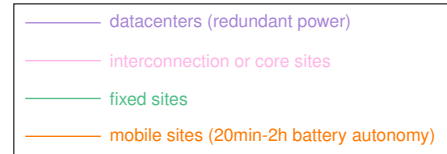
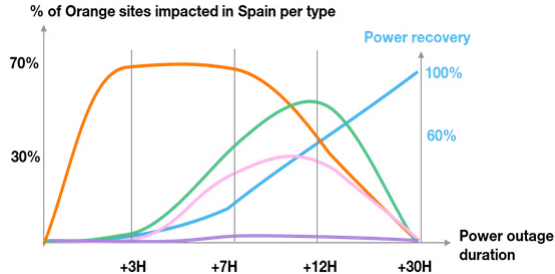
Août 2026 est le mois le plus chaud jamais mesuré dans le monde, Copernicus alerte sur des risques accrus pour l'Europe



Exogenous causes and consequences

Networks are interdependent

- Telecommunication networks require energy / Energy networks require transmission of information
- Transportation needed to access sites / Vehicles and infrastructure require connectivity



Impact of April 2025 power blackout in Spain on Orange sites, [Revue de l'Electricité et de l'Electronique 01/2026](#)

A network is a system of systems

Network resilience can be affected by many parameters

- hardware issues
- software issues (configuration, bug, ...)
- user traffic
- lifecycle management: updates, upgrades,
- weather conditions

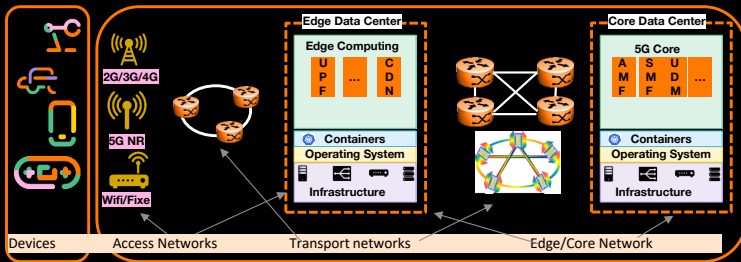
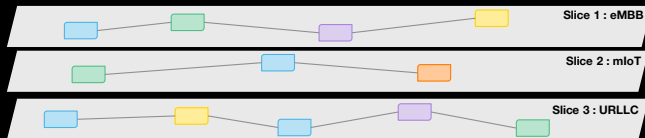
5G and beyond networks are now built different

- Virtualization with Network Function Virtualization (NFV) and Software-Defined Networking (SDN)
- Slicing
- Cloud native microservices

1



eMBB : enhances Mobile Broadband
mIoT : massive Internet of Things
URLLC : Ultra Reliable Low Latency Communication



Services
Orchestration



AI

Network
Functions
Orchestration

Network
Functions
Management

Infrastructure
Orchestration

[1] Rui Li, Bertrand Decocq, Anne Barros, Yiping Fang, Zhiguo Zeng. Complexity in 5G Network Applications and use cases. *31st European Safety and Reliability Conference*, Sep 2021, Angers, France. pp.3054-3061.

Modeling Formalisms: An End-to-End View

Non state-space models

- Fast and simple to solve
- Limited ability to capture dynamic behavior

Examples: RBDs, reliability graphs, fault trees

State-space models

- Capture state, time, and component dependencies
- May suffer from **state-space explosion**

Examples: CTMCs, Petri nets

Multi-level models

- Combine state-space and non state-space models
- Suitable for large and complex infrastructures

Also called hierarchical models

Key trade-off

Simple models are efficient but less expressive; dynamic models are more realistic but can become difficult to analyze as the number of states grows.

Toolbox

Non state-space models

Those methods can be **quickly** solved with the assumptions that the components are **statistically independent**, but they suffer from the **lack of performance in capturing the stochastic behavior** of such a system as they have no track over the system states. Traditional formalisms includes **Reliability Block Diagrams (RBDs)**, **Reliability graphs** and **Fault Trees**.

State-space models

those methods have the ability to incorporate statistical dependencies as well as dependencies on state and time, but they suffer from the analysis complexity and the state-explosion phenomena (the number of states increases exponentially which makes analysis harder and in some cases impossible with existing tools and methods). Those models can be:

- Markovian models: Discrete-time Markov chains (DTMCs), Continuous-Time Markov Chains (CTMCs), Markov Reward Model (MRM)
- non-markovian models: Petri Nets, Stochastic Reward Nets (SRNs)

Definitions

For a system to be resilient, it must first be operational. A crucial property is its reliability/availability, which strongly depends on its architecture, size, and components.

- **Reliability**: the probability $R(t)$ that an equipment works between 0 and t , with $R(0) = 1$.
- **Availability**: the probability $A(t)$ that an equipment is operating at t [mostly used in repairable systems].
- Other important notions:
 - **steady-state values, transients**
 - **sensitivity** (single points of failures)
 - **common-cause failures**
 - **resilience**
 - **size effects**



Applications (and associated mathematical tools)

- **Fixed networks:** End-to-end availability (graph theory, algebra)
- **Mobile networks:** Hand-over rate (mobility models, probabilistic geometry, calculus)
- **Data centers:** File and service availability (dealing with large numbers)
- **General studies:** Are usual assumptions safe? (non-Markovian methods, transients)
- **What happens for large systems?** Asymptotic expansions
- **New fields** are emerging in Safety and Reliability:
 - 5G/6G issues (latency)
 - Network virtualization
 - Quantum reliability (communications, computing)



Aim

- Find exact or good estimates of the availability of the system or service (analytical expressions a plus)
- Identify the key parameters of the problem under consideration
- Go beyond the Markov approximation, which is not always realistic and may overestimate reliability
- Provide a better understanding that can be applied in ensuing simulations and policies (digital twins, maintenance)



Selected research activities



Meshed networks [3]

The exact expression of the end-to-end availability between S and T can be obtained, not by using series-parallel simplifications, but from a generic solution for recursive families of graphs (917 terms)

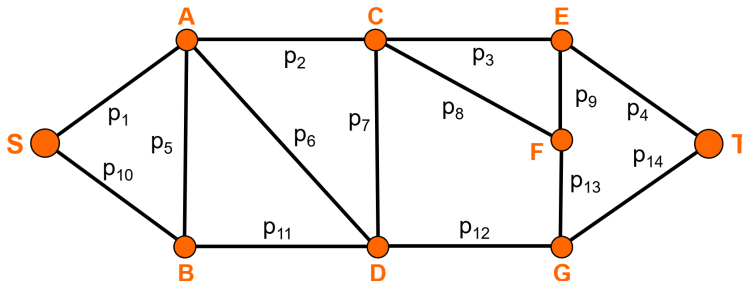


figure from [2]



Meshed networks [3]

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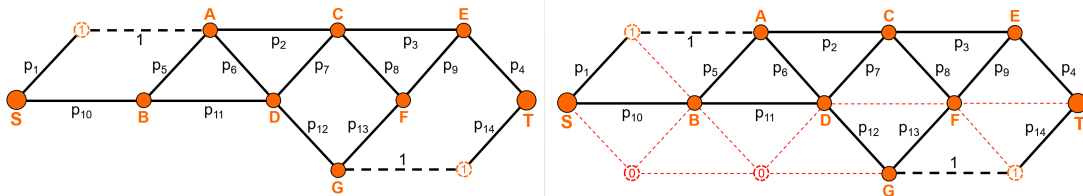
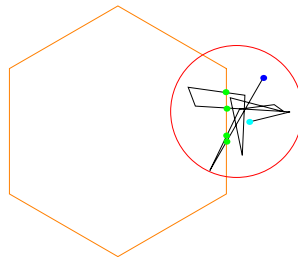
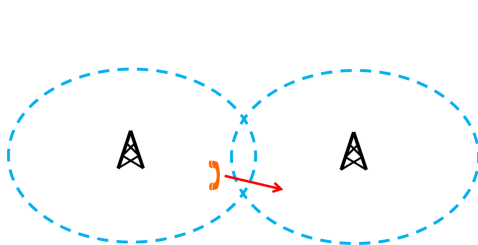


figure from [2]



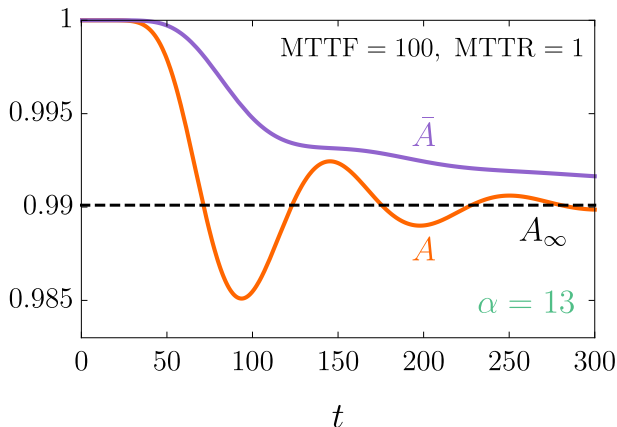
Hand-over rate and Random Waypoint model [4]



- What is the probability of leaving a cell?

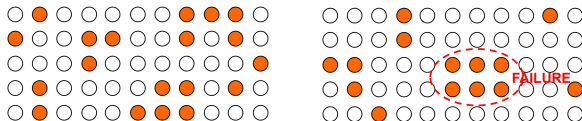


Importance of transients [5]



Instant and averaged availabilities $A(t)$ and $\bar{A}(t)$ for a gamma($\lambda, 13$) lifetime distribution and an exponential repair distribution

Two-dimensional (3,2)-out-of-(11,5):F system [6]



$$\begin{aligned}
 \mathcal{R} = & (1 - q)^6 \left(1 + 6q + 21q^2 + 56q^3 + 126q^4 + 252q^5 + 426q^6 + 576q^7 \right. \\
 & + 563q^8 + 205q^9 - 651q^{10} - 2111q^{11} - 3865q^{12} - 4912q^{13} - 3949q^{14} \\
 & - 111q^{15} + 5867q^{16} + 12046q^{17} + 15332q^{18} + 12047q^{19} + 1086q^{20} \\
 & - 14387q^{21} - 25509q^{22} - 24554q^{23} - 11602q^{24} + 7156q^{25} + 22184q^{26} \\
 & + 27253q^{27} + 18578q^{28} - 240q^{29} - 16433q^{30} - 19156q^{31} - 9994q^{32} \\
 & + 856q^{33} + 6078q^{34} + 5867q^{35} + 3723q^{36} + 731q^{37} - 1794q^{38} \\
 & - 1925q^{39} - 344q^{40} + 406q^{41} + 69q^{42} - 38q^{43} + 21q^{44} \\
 & \left. + 41q^{45} + 12q^{46} - 4q^{47} - 4q^{48} - q^{49} \right)
 \end{aligned}$$

where q is the unavailability of each element



Selected publications

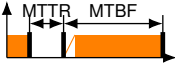
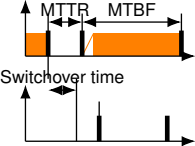
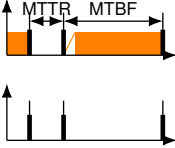
- [3] Christian Tanguy. “What is the probability of connecting two points?” In: *Journal of Physics A: Mathematical and Theoretical* 40.47 (2007), pp. 14099–14116.
- [4] Christian Tanguy. “Handover Rate in Wireless Networks and Random Waypoint Mobility Model: An Almost Simple Result”. In: 10th International Conference on Mathematical Methods in Reliability (Grenoble, France). 2017.
- [5] Christian Tanguy. “Transient Behaviour of Instantaneous and Average Availabilities in Non-Markovian Configuration”. In: *Safety and Reliability of Systems and Processes* (2022), pp. 185–194. DOI: [10.26408/srsp-2022-13](https://doi.org/10.26408/srsp-2022-13).
- [6] Christian Tanguy. “Impact of Common-Cause Failures on the Availability of Connected (r, s)-out-of-(m, n): F Systems”. In: *35th European Safety and Reliability Conference (ESREL 2025) and the 33rd Society for Risk Analysis Europe Conference (SRA-E 2025)*. 35th European Safety and Reliability Conference (ESREL 2025) (Stavanger, Norway). RESEARCH PUBLISHING, SINGAPORE, 2025. ISBN: 978-981-94-3281-3. DOI: [10.3850/978-981-94-3281-3-product](https://doi.org/10.3850/978-981-94-3281-3-product)

Selected publications (cont.)

- [22] Christian Tanguy. “When Considering the Asymptotic Availability Is Not a Safe Bet”. In: *Proceedings of the 30th European Safety and Reliability Conference and the 15th Probabilistic Safety Assessment and Management Conference*. (Venice, Italy). Research publishing, singapore, 2020, pp. 3091–3098. ISBN: 978-981-14-8593-0.
- [23] Christian Tanguy. “Shannon Entropy and Its Variants for the Maintenance of Large Meshed System Architectures”. In: *Proceedings of the 13th IMA International Conference on Modelling in Industrial Maintenance and Reliability 2025*. 13th IMA International Conference on Modelling in Industrial Maintenance and Reliability (MIMAR) (Nancy, France). Institute of Mathematics and its Applications, 2025. DOI: [10.19124/ima.2025.01](https://doi.org/10.19124/ima.2025.01).

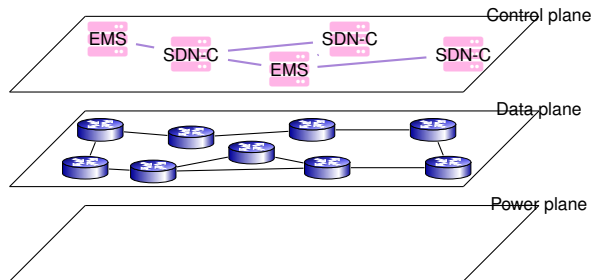


Typical resilience mechanisms for telcos

		Scheme	Backup ratio	Failure-Repair Sequence	Interruption Time
	App. Server	Self Healing	0		MTTR
	5G User Plane	N:B Shared Protection	B/N		switchover time
	5G Control Plane	1+1 active/ac- tive protection	1		0

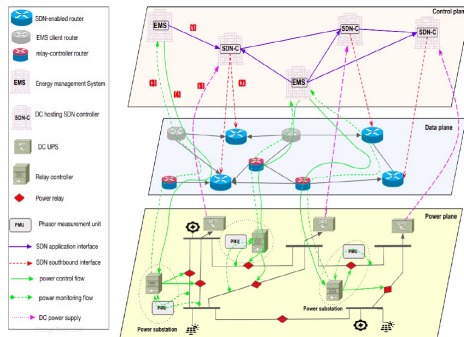
Cross-domain resilience in Cloud-native, Critical Cyber Physical Systems Networks

Results: Model for the propagation of failures between **interdependent** critical infrastructures and analyze its impact on overall availability. Optimize the placement of virtualized functions across multiple sites based on availability and latency constraints. Propose an architecture for exchanging availability information to plan maintenance operations.



Supervised Learning of virtualization infrastructure availability

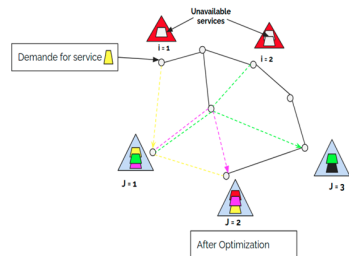
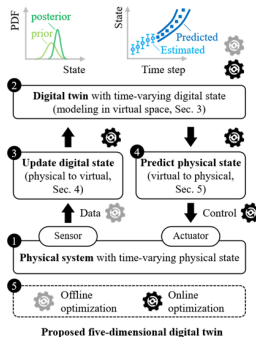
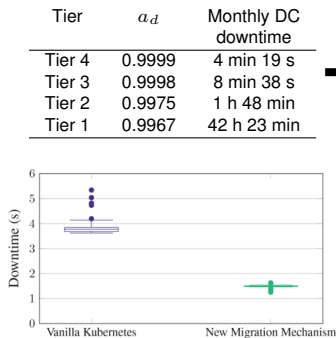
<https://theses.hal.science/tel-04713044>



From Khaled Sayad et al [21]

Supervised Learning of virtualization infrastructure availability

Objective: Model the failure statistics of infrastructure components and develop a supervised learning approach for the model parameters-aimed at compressing measurements from infrastructure sensors for a Digital Twin-while retaining only information regarding rare events such as failures, resilience, and the return to normal operations.



Selected publications

- [7] Benoît Lemoine. “Energy Efficiency of N:1 Protection Setups with Kubernetes Horizontal Pod Autoscaler”. In: *2022 25th Conference on Innovation in Clouds, Internet and Networks (ICIN)*. 2022 25th Conference on Innovation in Clouds, Internet and Networks (ICIN). Mar. 2022, pp. 126–130. DOI: [10.1109/ICIN53892.2022.9758129](https://doi.org/10.1109/ICIN53892.2022.9758129).
- [8] Benoît Lemoine. “Kubernetes Horizontal Pod Autoscaler Enhanced Design for Protected Workloads”. In: *2024 IEEE Symposium on Computers and Communications (ISCC)*. 2024 IEEE Symposium on Computers and Communications (ISCC). June 2024, pp. 1–7. DOI: [10.1109/ISCC61673.2024.10733692](https://doi.org/10.1109/ISCC61673.2024.10733692).
- [18] Khaled Sayad et al. *Availability Modeling and Analysis of Cloud-Native, Interdependent Cyber-Physical*. Feb. 16, 2024. DOI: [10.2139/ssrn.4728817](https://doi.org/10.2139/ssrn.4728817). Social Science Research Network: [4728817](https://ssrn.com/abstract=4728817). Pre-published.



Selected publications (cont.)

- [19] Khaled Sayad. “Cross-Domain Resilience in Cloud-native, Critical Cyber-Physical Systems Networks : Availability Modeling, Analysis, and Optimization of Critical Services Provisioning”. These de doctorat. université Paris-Saclay, Apr. 10, 2024. URL: <https://theses.fr/2024UPAST028> (visited on 07/17/2026).
- [20] Khaled Sayad et al. “Interdependency-Aware Resource Allocation for High Availability of 5G-enabled Critical Infrastructures Services”. In: *32nd European Safety and Reliability Conference, ESREL 2022*. Dublin, Ireland, Aug. 2022. DOI: [10.3850/978-981-18-5183-4_S28-03-640-cd](https://doi.org/10.3850/978-981-18-5183-4_S28-03-640-cd).
- [21] Khaled Sayad and Benoît Lemoine. “Towards Cross-domain Resilience in SDN-enabled Smart Power Grids: Enabling Information Sharing through Dataspaces”. In: *2023 IEEE International Conference on Omni-layer Intelligent Systems (COINS)*. 2023 IEEE International Conference on Omni-layer Intelligent Systems (COINS). July 2023, pp. 1–6. DOI: [10.1109/COINS57856.2023.10189319](https://doi.org/10.1109/COINS57856.2023.10189319).



From availability assessment to automation

Availability can be evaluated using several **well-established approaches**.

Analytical
models

Markov
models

RBD / Fault Trees
reliability models

Experimental
evaluation

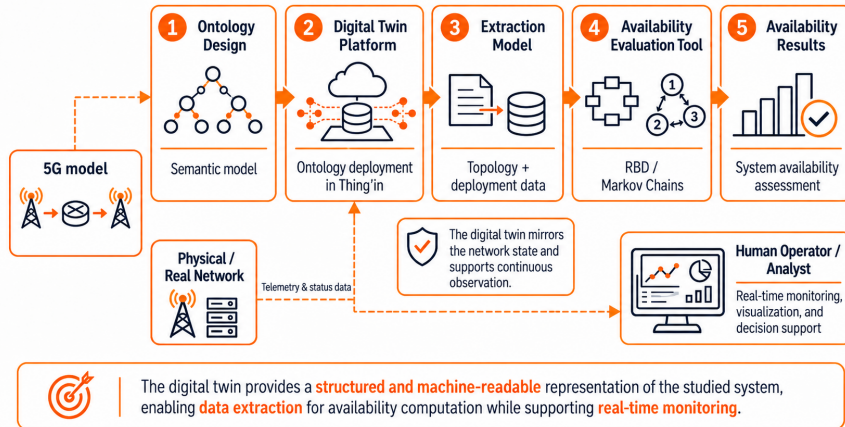
Practical challenge

The methods themselves are mature, but building a complete end-to-end assessment often requires manual effort for **model construction • scenario configuration • failure injection • data collection • analysis**.



Digital Twins provide a promising foundation for making availability assessment **systematic, repeatable and automatable**.

Digital twin technology



Use case: TMForum : Living Network – Phase III Catalyst

The studied use case is composed of two service chains: a **User Plane Chain** and a **Control Plane Chain**.

User Plane Chain



Control Plane Chain



Modeling assumptions: each **NF/VNF** is decomposed into **3 microservices**, and each microservice is instantiated with **2 replicas**.



Ontology Design for the 5G Digital Twin

The ontology provides a **semantic representation** of the main entities, relationships and attributes of the virtualized 5G system.

Ontology classes

VNF ● Microservice ● Replica ● Pod ● Server

Object properties

hasMicroservice: VNF → Microservice

hasReplica: Microservice → Replica

deployedIn: Replica → Pod

hostedOn: Pod → Server

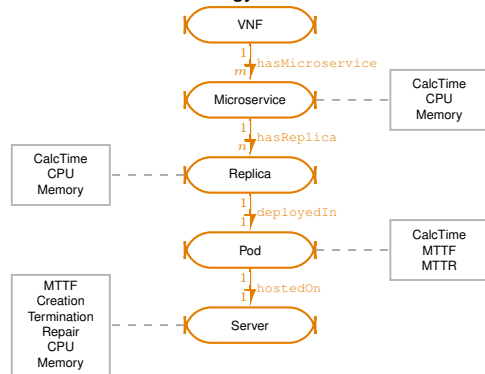
Data properties

Microservice / Replica: CalcTime, CPU, Memory

Pod: MTTF, MTTR, CalcTime

Server: MTTF, MTTR, Creation, Termination, CPU, Memory

Ontology schema



The ontology captures both the **logical service structure** and its **deployment infrastructure**

Other actions are attached to a chair...

Risk and Resilience of Complex Systems Chair of Centrale Supélec, led by Anne Barros https://r3.centralesupelec.fr/en/Chair_RRCS



Scientific Project 2024-2029:

- Axis 1: System Modeling for Risk, Resilience, and Reliability
- Axis 2: Decision Support under Uncertainty



Partners



Bibliography

- [24] Ikram Temericht et al. "Optimizing 5G Network Availability: Comparison Between Reliability Block Dia-grams and Markov Chains Approaches". In: (2025). URL: <https://rpsonline.com.sg/proceedings/esrel-sra-e2025/pdf/ESREL-SRA-E2025-P5793.pdf> (visited on 07/17/2026).



谢谢

Thank you

Merci

*"The greatest glory
of living lies not
in never falling but
in rising every time
you fall."*

Nelson Mandela

